Software Requirements Specification

for

CollabCupid: Post Ideas. Find Partners. Build for Fun.

Table of Contents

Table of Contents
Revision History
1. Introduction
1.1 Purpose
1.2 Intended Audience
1.3 Product Scope
1.4 References
2. Overall Description
2.1 Product Perspective
2.2 Product Functions
2.3 User Classes and Characteristics
2.4 Operating Environment
2.5 Design and Implementation Constraints
2.6 Assumptions and Dependencies
3. External Interface Requirements
3.1 User Interfaces
3.2 Software Interfaces
3.3 Communications Interfaces
4. Analysis Models
4.1 Project Owner Use case diagram
4.2 Collaborator Use case diagram
4.3 Project Owner Class Diagram
4.4 Collaborator Class DIagram
4.5 ER diagram
5. System Features
5.1 User Accounts
5.2 Project discovery
5.3 Collaborator Discovery
5.4 Repository Management
5.5 Messaging and Notifications
5.6 Recommendation System
6. Other Nonfunctional Requirements
6.1 Performance Requirements
6.2 Safety Requirements
6.3 Security Requirements
6.4 Software Quality Attributes
6.5 Business Rules
7. Other Requirements
Appendix A: Glossary

Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification is to provide a clear, documented model of the requirements for the CollabCupid system. This document provides the top-level use cases for students collaborating on projects through an online platform. The system includes the student subsystem as well as the administrator subsystem.
CollabCupid provides a platform where students can post project ideas and find collaborators with similar interests and skills. It is implemented as an internet-based application that enables students to build teams and work together on academic and personal projects.
The system helps students easily find suitable teammates, encourages collaboration, and promotes innovation through effective project development.

1.2 Intended Audience

The website specifically targets ambitious university and high school students who want to build the real-world experience outside the traditional classroom setting. It caters to project owners who have innovative concepts but lack the specific skills to execute them alone, as well as collaborators looking to join existing teams to sharpen their portfolios in coding, design, or marketing. By bringing together interdisciplinary self-starters, the site serves as a digital incubator for hackathon teams, peer-to-peer networking, and early-stage startup co-founders. Ultimately, it provides a low-pressure, accessible space for both introverted students and global innovators to cross geographic boundaries, connect over shared tech interests, and turn theoretical knowledge into tangible, collaborative products.

1.3 Product Scope

CollabCupid provides a platform for students to share project ideas and collaborate with other students on academic, personal, research, and innovation-based projects. It serves as an online platform that brings together students with project ideas and those who possess the skills and interest required to contribute. The system simplifies the process of finding suitable collaborators and forming effective project teams.

The platform provides students with the ability to create profiles, publish project ideas, browse projects posted by others, and send requests to join project teams. Project creators can review applications, accept or reject collaborators, and manage their project teams. The system also enables communication between team members, allowing them to coordinate their work and share project updates efficiently.

One of the major advantages of the system is that it removes the difficulty of finding teammates beyond one's immediate circle of friends or classmates. By connecting students from different departments, years of study, and technical backgrounds, the platform encourages interdisciplinary collaboration, knowledge sharing, and innovation. It provides equal opportunities for students to participate in projects that match their interests and skills.

The vision of CollabCupid is to create a collaborative student community where ideas can be transformed into successful projects through teamwork and effective communication. The system aims to provide a secure, user-friendly, and accessible environment that supports project-based learning, hackathons, competitions, research activities, and other collaborative initiatives while encouraging creativity and professional growth among students.

1.4 References

[1] Collaborative platform for university and college Students Project
Authors: Madhavi Kulkarni, Aditya Jadhav, Mohanlal Koli, Abhijeet Magar, Aditya Navale
Date: January 2025
[2] Success with your education research project
Author: John G. Sharp
Date: 2009
[3] Online Group Projects in higher education
Author: Helen Donelan, Karen Kear
Date: March 2023
[4] An Interactive Web-Based Platform for Building Academic Collaboration
Author: Joshua D. Brown, Nadja Kadom, Elizabeth A. Krupinski
Date: May 2024

2. Overall Description

2.1 Product Perspective

The system includes the student subsystem as well as the administrator subsystem. CollabCupid provides a platform for students to post project ideas and collaborate with others in an efficient and organized manner. It enables students to discover projects, find suitable teammates, and work together without geographical or departmental limitations. The system serves as a one-stop platform for student collaboration, promoting teamwork, innovation, and effective project management.

2.2 Product Functions

Enlisted below are all the major functions supported by the collaboration platform along with the user classes.

- **Register:** for project owners and collaborators
- **Login:** for project owners and collaborators
- **Logout:** for project owners and collaborators
- **View Account Details:** for project owners and collaborators
- **Edit Account Details:** for project owners and collaborators
- **Search collaborator:** for project owners
- **View collaborator:** for project owners
- **Search projects:** for collaborators
- **View project details:** for collaborators
- **Add project to profile:** for collaborators
- **Add collaborator to project:** for project owners
- **Remove collaborator from project:** for project owners
- **Edit project details:** for project owner
- **Add project files to repository:** for project owners and collaborators

2.3 User Classes and Characteristics

Project Owner - A verified student who creates and manages project postings on the platform. The Project Owner can register, log in, create and edit project posts, specify the required skills and team size, view collaboration requests, accept or reject applicants, communicate with team members, and manage the project until its completion.

Collaborator - A verified student who uses the platform to discover projects that match their interests and skills. The Collaborator can register, log in, create and update their profile, browse and search for projects, send requests to join project teams, communicate with project owners and teammates after acceptance, and participate in collaborative project development.

2.4 Operating Environment

CollabCupid is available as a responsive, single-page web application. It can be accessed using modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari on Windows, macOS, and Linux operating systems, and the interface adapts to tablets and smartphones through the same browser-based experience rather than a separate native app. An active internet connection is required for users to access the platform and its features. The system uses Google Firebase as its cloud-based backend to store user profiles, project details, and collaboration information.

2.5 Design and Implementation Constraints

The system is designed exclusively for students and requires users to register using a valid email address. A project owner can create and manage multiple project posts, while collaborators can apply only through the provided application feature. Project owners have complete control over accepting or rejecting collaboration requests. The system depends on a stable internet connection for real-time access to project listings, notifications, and communication features. Since the application relies on a centralized database, server availability is essential for uninterrupted operation. The messaging feature is limited to project members. The system is designed for academic and collaborative purposes only. Additional features and third-party integrations may be incorporated in future versions without significantly modifying the existing system architecture. To ensure ease of maintenance and future scalability, the system is designed using a modular architecture. Each major component, including user management, project management, collaboration requests, and communication, functions independently while integrating seamlessly with the rest of the system. This approach simplifies system updates, feature enhancements, and maintenance without significantly affecting existing functionalities.

2.6 Assumptions and Dependencies

The system assumes that users have access to a stable internet connection and a supported web browser or mobile device. It also assumes that users register using valid institutional email addresses. The application depends on a centralized database to store user and project information. The availability of the server and network services is essential for the proper functioning of the system. The system assumes that users provide accurate profile information, including their skills and interests, to facilitate effective collaboration. It also assumes that project owners and collaborators will use the platform responsibly and follow the guidelines for project creation and participation.

3. External Interface Requirements

3.1 User Interfaces

-> <u>Home Page</u>:
The Home Page serves as the landing page of the application.
It contains:

- Login button
- Register button
- Search bar
- Filter button
- Project recommendations
- Recently created projects
- Projects with open collaboration positions
- Navigation drawer (top-left)

The navigation drawer contains:

- View Profile
- My Projects
- Repository
- Chats
- Notifications
- Settings
- Logout

Projects displayed on the Home Page are automatically filtered to include only projects that still have available positions.

-> <u>Login Page</u>

Allows registered users to access their account.
Fields

- Email
- Password

Buttons

- Login
- Forgot Password
- Register

<u>-> Registration Page</u>
Allows new users to create an account.
Fields

- Name
- Institution
- Email
- Password
- Confirm Password
- Skills
- Interests

-> User Profile

Displays

- Profile picture
- About section
- Skills
- Technologies known
- Previous projects
- Repository
- Current collaborations
- Contact information (optional)

Users may edit their profile information.

-> Project Details Page

Selecting a project opens its project profile.
It displays

- Project Title
- Description
- Category
- Required Skills
- Open Positions
- Project Owner
- Repository Status
- Team Members

Buttons

- Request to Join
- Contact Project Owner

The user selects the position they are applying for while sending the request.
When every required position has been filled, the project is automatically hidden from search results and the Home Page.

-> Search Page

The search interface supports searching for

- Projects
- Collaborators

Filters include

- Skill
- Technology
- Category
- Institution

-> Repository Page

Each project has a dedicated repository.
Features

- Upload files
- Download files
- Edit code
- Save changes (creates a new file version)
- View version history

The repository is accessible only to project members.

-> Chat Interface
Supports
- One-to-one messaging
- Project group chat
- Typing indicators
- Online status

Messages update in real time without refreshing the page.

3.2 Software Interfaces

The system consists of the following software components.

Frontend
- React with TypeScript
- Tailwind CSS
- Motion (Framer Motion)
- Lucide React Icons
- Monaco Editor

Responsibilities
- Responsive UI
- State management
- Repository editor
- Search filtering
- Chat interface

Backend
CollabCupid has no custom backend server; the frontend talks directly to Firebase's managed services, which provide
- Authentication
- User management
- Project management
- Collaboration requests
- Repository management
- Notifications
- Chat

Authorization
- Enforced via Firestore Security Rules, evaluated on Google's servers

Database
Cloud Firestore (NoSQL document database)
Stores
- Users
- Projects
- Collaboration Requests
- Conversations & Messages
- Notifications
- Repository File Metadata

Repository Storage
Project files are stored using
- Firebase Storage

Each uploaded file is associated with
- Repository
- Project
- Upload Date
- Version
- Contributor

3.3 Communications Interfaces

Communication between users is established directly between the client and Firebase's managed services, secured over HTTPS/SSL via Firestore's listeners.
The system supports

Collaboration Requests
Project owners invite collaborators.
Collaborators request to join projects.
Each request contains
- Project
- Requested Position
- Status
- Timestamp

Possible states
- Pending
- Accepted
- Rejected

Upon acceptance
- collaborator added
- repository unlocked
- project chat unlocked

Real-Time Chat
Implemented using
Cloud Firestore real-time listeners (onSnapshot)
Supports
- Direct Messages
- Project Group Chats
- Typing Indicators
- Online Presence

Notifications
Users receive notifications when
- Collaboration request received
- Request accepted
- Request rejected
- Added to project
- Removed from project
- Repository updated
- New message received

4. Analysis Models

4.1 Project Owner Use Case Diagram

This use case diagram represents the interactions between the Project Owner and the CollabCupid system. The Project Owner can post and edit project ideas, search and communicate with collaborators, and review collaboration applications. The owner can accept or reject applicants and, after completing the project, push it to the project repository.

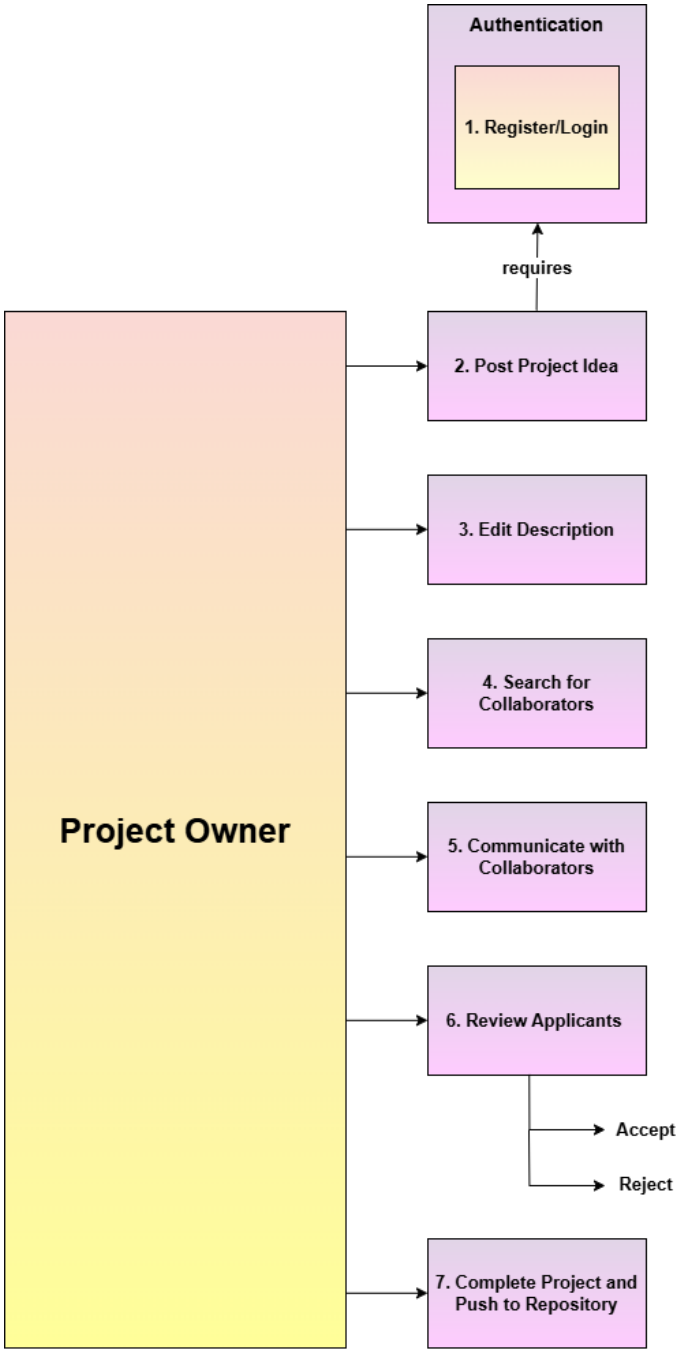


Fig 4.1 Project owner use case diagram

4.2 Collaborator Use Case Diagram

The Collaborator can register/login, edit their profile and skills, and search for suitable projects. They can request to join projects and, upon acceptance, access the project's secure chat for team communication. After project completion, the Collaborator can link the project repository to their profile to showcase their contribution**.**

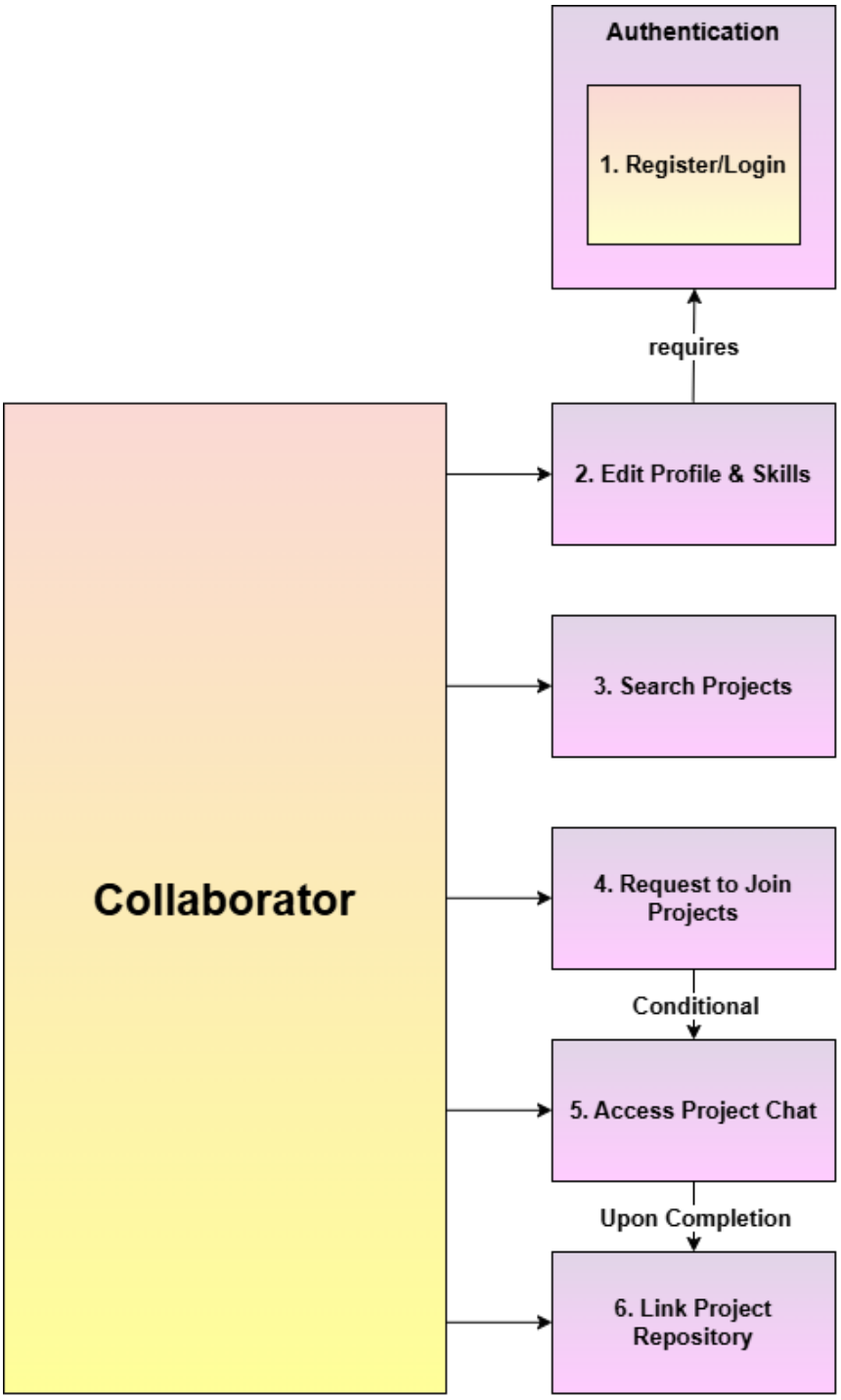


Fig 4.2 Collaborator use case diagram

4.3 Project Owner Class Diagram

The class diagram represents the ProjectOwner class and its relationships with the Project and CollaborationRequest classes. The ProjectOwner can register, log in, manage account details, create and edit projects, search and manage collaborators, and communicate with team members. A ProjectOwner owns one or more projects and can review collaboration requests received for those projects, while each Project is associated with its collaboration requests.

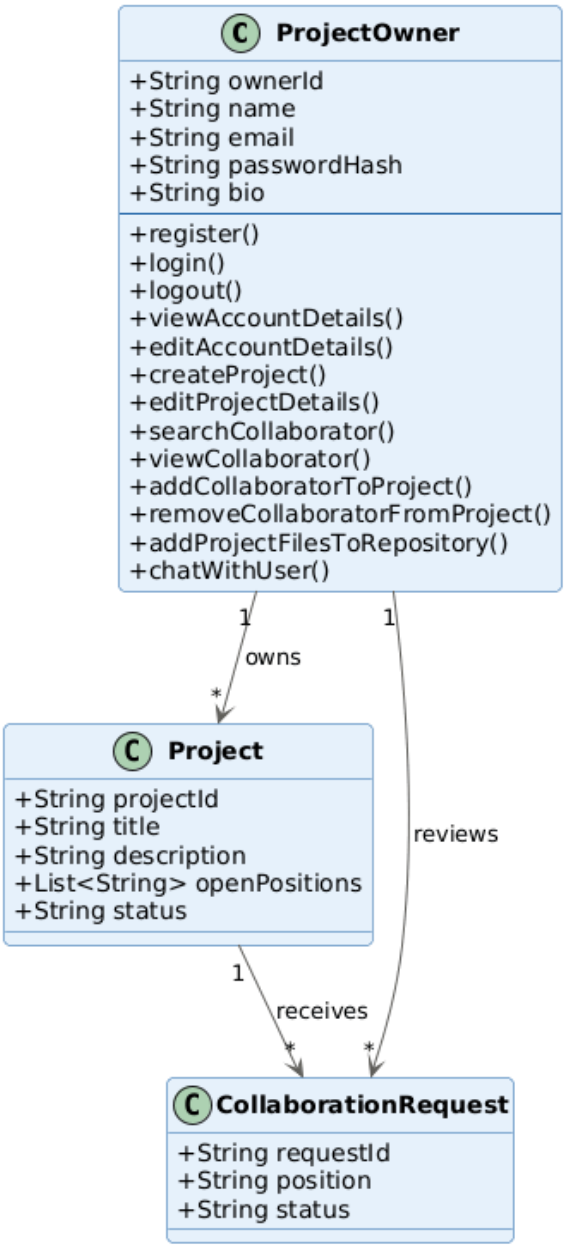


Fig 4.3 Project owner class diagram

4.4 Collaborator Class Diagram

The class diagram represents the Collaborator class and its relationships with the Project and CollaborationRequest classes. The Collaborator can manage their account and skills, search and view projects, request to join projects, add projects to their profile, and communicate with other users. A Collaborator can join multiple projects and send collaboration requests targeting specific projects, while each request is associated with one project.

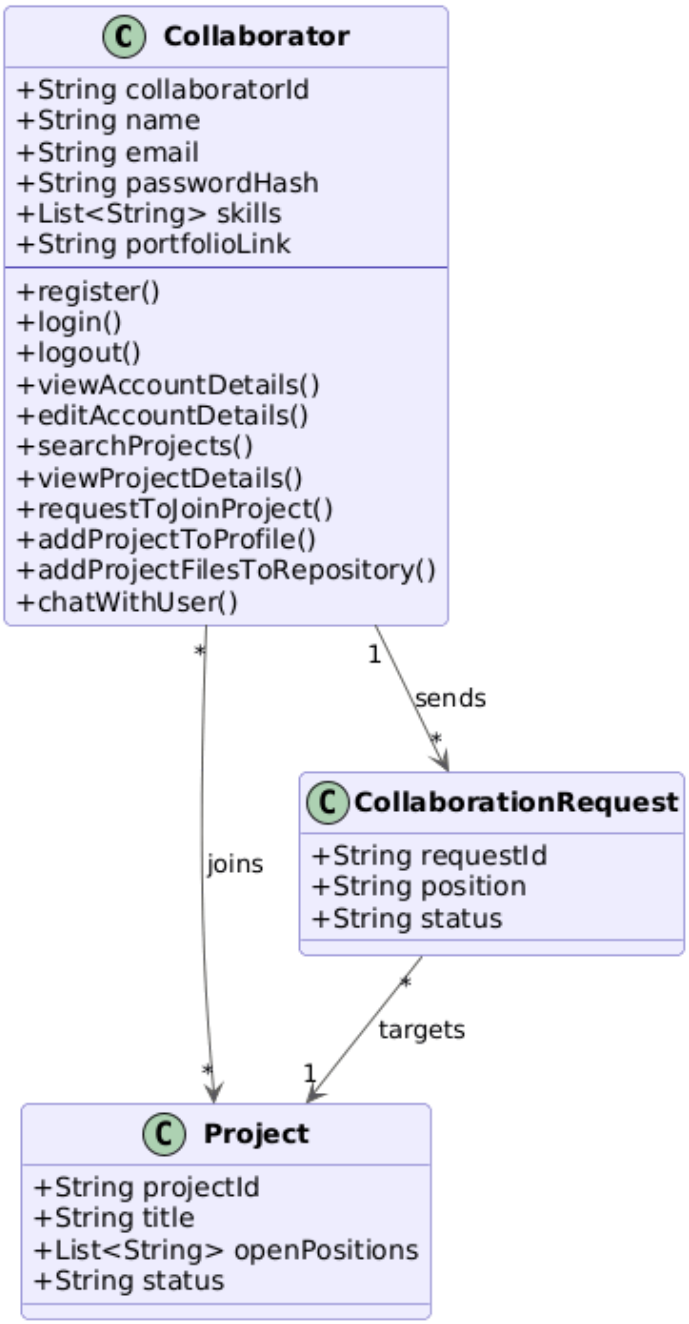


Fig 4.4 Collaborator class diagram

4.5 ER Diagram

The ER diagram represents the database structure of CollabCupid, including users, projects, collaboration requests, project members, repository files, conversations, and messages. It defines the relationships between these entities, such as users owning projects, sending collaboration requests, joining projects, and participating in conversations. The diagram also shows how project repositories store files and how conversations contain messages exchanged between project members**.**

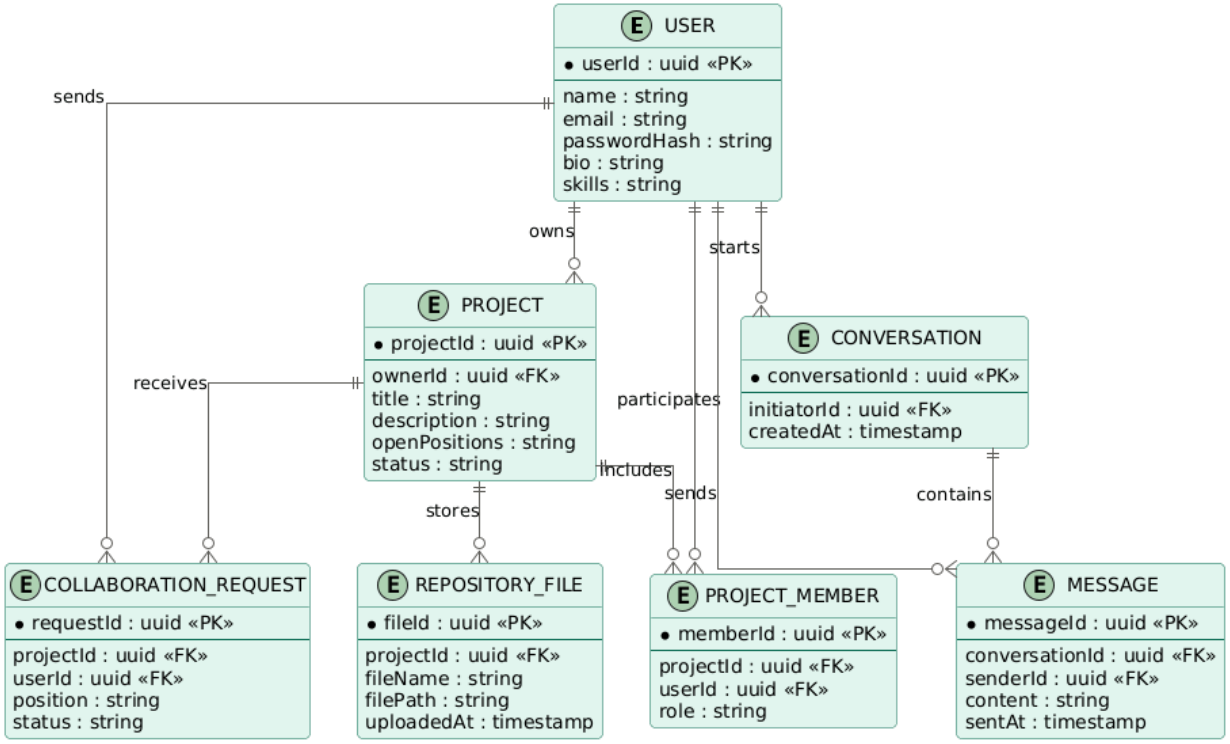


Fig 4.5 ER Diagram

5. System Features

5.1 User Accounts

5.1.1 Description and Priority

The User Accounts feature provides secure authentication and profile management for all users of the CollabCupid platform. Every user must create an account before they can create projects, search for collaborators, send collaboration requests, participate in chats, or contribute to project repositories. Users who are not logged in are limited to browsing the Home Page and viewing public project listings without interacting with the platform.

Each user account stores personal information including the user's name, institutional email address, password, profile picture, educational institution, technical skills, areas of interest, previous projects, portfolio repository, and a short biography. The information stored in the profile enables project owners to evaluate potential collaborators and helps the recommendation system suggest suitable projects and teammates.

Users may update their account information at any time after logging in. Sensitive information such as passwords is securely encrypted and is never displayed in plain text. Authentication is handled through Firebase Authentication, while user profile information is maintained within the application database.

Since every other module of the system depends upon authenticated users, the User Accounts module is considered one of the most critical components of the platform.

Priority Level: High

5.1.2 Stimulus/Response Sequences

A new user visits the CollabCupid Home Page and selects the Register option located in the navigation bar. The system displays the registration form where the user enters the required information such as name, institutional email address, password, educational institution, technical skills, and areas of interest. After validating all mandatory fields and verifying that the email address has not already been registered, the system creates a new account and prompts the user to verify their email address. Once verification is complete, the user is redirected to the Login page.

An existing user selects the Login option from the Home Page and enters their registered email address and password. If the credentials are valid, the system authenticates the user, establishes a secure session, retrieves the user's profile information from the database, and redirects the user to the personalized Home Page displaying recommended projects.

At any point after logging in, the user may access the navigation drawer and choose View Profile to display their account details, including personal information, skills, repositories, previous projects, and current collaborations.

The user may also select Edit Profile, modify one or more profile fields, and click Save Changes. The system validates the updated information before storing it in the database and immediately reflects the changes throughout the platform.

When the user chooses Logout, the system terminates the authenticated session, clears all session tokens, and redirects the user to the Home Page. Any attempt to access authenticated pages after logout redirects the user back to the Login page.

5.1.3 Functional Requirements

REQ-1: Register
User: Project Owners and Collaborators
Input:
In the Registration page

- Enter Full Name
- Enter Institutional Email Address
- Enter Password
- Confirm Password
- Select Educational Institution
- Enter Skills
- Enter Areas of Interest
- Upload Profile Picture (Optional)
- Click Register

Output:

- User account is successfully created.
- Verification email is sent to the registered email address.
- User is redirected to the Login page after successful verification.

Alternative flow(s):

- If the email address already exists, the system displays:

"An account with this email already exists."

- If the password does not satisfy the required security policy, the system prompts the user to create a stronger password.
- If mandatory fields are left empty, the system highlights the missing fields and prevents registration.
- If the password and confirmation password do not match, the system asks the user to re-enter both passwords.

REQ-2: Login
User: Project Owners and Collaborators
Input:
In the Login page

- Enter Registered Email Address
- Enter Password
- Click Login

Output:

- User is successfully authenticated.
- User session is established.
- Personalized Home Page displaying recommended projects is displayed.

Alternative flow(s):

- Invalid email address.
- Incorrect password.
- Email address not verified.
- User account temporarily disabled by administrator.

In all such cases, an appropriate error message is displayed and the user is requested to re-enter valid credentials.

REQ-3: View Profile
User: Project Owners and Collaborators
Input:

- Open Navigation Menu.
- Click View Profile.

Output:
The system displays the complete user profile containing:

- Name
- Profile Picture
- Educational Institution
- About Section
- Skills
- Areas of Interest
- Previous Projects
- Current Projects
- Repository
- Portfolio Links
- Collaboration History

Alternative flow(s):

- If the profile information cannot be retrieved due to a server error, the system displays: "Unable to load profile. Please try again later."

REQ-4: Edit Profile
User: Project Owners and Collaborators
Input:
From the Profile page

- Click Edit Profile
- Modify one or more profile fields:
    - Name
    - Profile Picture
    - Biography
    - Skills
    - Areas of Interest
    - Educational Institution
    - Portfolio Links
- Click Save Changes

Output:

- Updated profile information is validated.
- Changes are saved successfully.
- Updated information is immediately visible throughout the platform.

Alternative flow(s):

- Invalid email format.
- Unsupported profile picture format.
- Profile picture exceeds maximum file size.
- Mandatory fields left blank.
- Database update failure.

In each case, the system displays an appropriate validation message and prevents saving until the issue is resolved.

REQ-5: Logout
User: Project Owners and Collaborators
Input:

- Open Navigation Menu.
- Click Logout.

Output:

- User session is terminated.
- Authentication token is invalidated.
- User is redirected to the Home Page.
- Protected pages become inaccessible until the user logs in again.

Alternative flow(s):

- If a temporary network interruption prevents immediate logout, the system retries the logout request and informs the user if the operation cannot be completed. Regardless of the network status, locally stored authentication tokens are cleared to prevent unauthorized access.

REQ-6: Change Password
User: Project Owners and Collaborators
Input:
From the Account Settings page

- Enter Current Password
- Enter New Password
- Confirm New Password
- Click Update Password

Output:

- Password is successfully updated.
- User receives a confirmation email notifying them of the password change.

Alternative flow(s):

- Current password is incorrect.
- New password does not satisfy password policy.
- New password matches the current password.
- Confirmation password does not match.

The system displays an appropriate error message and requests the user to correct the input.

REQ-7: Forgot Password
User: Project Owners and Collaborators
Input:
From the Login page

- Click Forgot Password
- Enter Registered Email Address
- Click Send Reset Link

Output:

- Password reset email is sent to the registered email address.
- User can securely create a new password using the reset link.

Alternative flow(s):

- Email address is not registered.
- User account has been disabled.
- Email service temporarily unavailable.

In these situations, the system advises the user to try again later or contact support.

5.2 Project Discovery

5.2.1 Description and Priority

The Project Discovery feature enables users to browse, search, and explore projects available on the CollabCupid platform. The Home Page displays projects that are actively recruiting collaborators based on the user's interests and skills. Users can search for projects using keywords or apply filters such as technology stack, project category, institution, required skills, and available roles. Selecting a project displays its complete details, including the project description, required collaborators, technologies used, repository status, current team members, and open positions. Users interested in contributing may submit a collaboration request for a specific role.
Projects whose collaboration positions have all been filled are automatically removed from the Home Page and search results until a new position becomes available.
Priority Level: High

5.2.2 Stimulus/Response Sequences

After logging into the system, the user is presented with the Home Page displaying recommended projects with open collaboration positions. The user may enter keywords into the search bar or apply filters to narrow down the displayed projects. The system searches the project database and displays projects matching the specified criteria.
The user selects a project from the search results or Home Page to view its details. The Project Details page displays information such as project description, required technologies, available roles, current collaborators, and repository information.
If the user wishes to contribute to the project, the **Request to Join** button is selected. The user specifies the desired role and optionally enters a brief introduction before submitting the request. The project owner receives a notification regarding the collaboration request.

5.2.3 Functional Requirements

REQ-1: Search Projects
User: Project Owners and Collaborators
Input:
In the Home Page

- Enter project name or keywords in the search bar

or

- Apply one or more filters:
    - Technology
    - Skill
    - Category
    - Institution

Output:
Displays a list of projects matching the search criteria.
Alternative flow(s):

- If no matching projects are found, the system displays:

- "No projects found."
- "Try modifying your search criteria."

REQ-2: View Project Details
User: Project Owners and Collaborators
Input:

- Select a project from the Home Page or search results.

Output:
Displays:

- Project Title
- Description
- Technologies Used
- Required Skills
- Available Positions
- Current Collaborators
- Project Owner
- Repository Status
- Request to Join button

Alternative flow(s**):**

- If the selected project has been deleted or recruitment has closed, the system informs the user that the project is no longer available.

REQ-3: Filter Projects
User: Project Owners and Collaborators
Input:
Select one or more filters from the filter panel.
Output:
The displayed project list is updated according to the selected filters.
Alternative flow(s):

- If no projects satisfy the selected filters, an appropriate message is displayed and the user may clear the filters.

REQ-4: Request to Join Project
User: Collaborators
Input:
In the Project Details page

- Click Request to Join
- Select desired project role
- Enter an optional introduction message
- Click Submit

Output:
A collaboration request is sent to the project owner, and the request status is marked as Pending.
Alternative flow(s):

- User has already submitted a request.
- User is already a collaborator in the project.
- The selected role has already been filled.
- The project owner has disabled collaboration requests.

5.3 Collaborator Discovery

5.3.1 Description and Priority

The Collaborator Discovery feature enables users to search for potential teammates based on their technical skills, educational institution, interests, previous projects, and areas of expertise. Users can view public profiles to evaluate a collaborator's experience before inviting them to join a project. This feature assists project owners in assembling balanced teams while allowing users to showcase their abilities and previous work.
The search functionality supports keyword-based searching as well as filtering by programming languages, frameworks, project domains, and educational institutions.
Priority Level: High

5.3.2 Stimulus/Response Sequences

The user selects the collaborator search option from the Home Page and enters keywords or applies filters such as skills, institution, programming language, or areas of expertise. The system retrieves and displays matching user profiles.
The user selects a collaborator profile to view detailed information including biography, technical skills, previous projects, repositories, and current collaborations.
If the project owner wishes to recruit the user, the Invite Collaborator button is selected. The project owner chooses one of their projects and specifies the role for which the collaborator is being invited. The selected user receives a collaboration invitation through the notification system.

5.3.3 Functional Requirements

REQ-1: Search Collaborators
User: Project Owners and Collaborators
Input:
In the Collaborator Search page

- Enter collaborator name or keywords

or
Apply filters:

- Skills
- Programming Languages
- Technologies
- Institution
- Areas of Interest

Output:
Displays a list of collaborator profiles matching the search criteria.
Alternative flow(s):

- If no matching users are found, the system displays:
    - "No collaborators found."
    - "Try changing your search filters."

REQ-2: View Collaborator Profile

User: Project Owners and Collaborators
Input:
Select a collaborator from the search results.
Output:
Displays:

- Profile Picture
- Name
- Institution
- Biography
- Skills
- Areas of Interest
- Previous Projects
- Repository
- Current Collaborations
- Contact Button

Alternative flow(s):

- If the collaborator profile is unavailable or has been deleted, an appropriate message is displayed.

REQ-3: Invite Collaborator
User: Project Owners
Input:
In the Collaborator Profile page

- Click Invite Collaborator
- Select one of the owned projects
- Select project role
- Enter an optional invitation message
- Click Send Invitation

Output:
A collaboration invitation is sent to the selected user, and its status is recorded as Pending.
Alternative flow(s):

- User is already a collaborator in the selected project.
- User has already been invited.
- The selected project has no open positions.
- The project owner has no projects available for invitation.

REQ-4: Contact Collaborator
User: Project Owners and Collaborators
Input:
From the Collaborator Profile page

- Click Chat
- Type message
- Click Send

Output:
A direct chat conversation is created if one does not already exist, and the message is delivered to the selected collaborator.
Alternative flow(s):

- Recipient account is inactive.
- Temporary network failure prevents message delivery.

- The system stores the message and retries delivery once connectivity is restored.

5.4 Repository Management

5.4.1 Description and Priority

The Repository Management feature provides a centralized location for storing, organizing, and maintaining all files related to a project. Each project is automatically assigned a repository upon creation, enabling collaborators to upload source code, documentation, presentations, datasets, and other project resources. Repository access is restricted to the project owner and authorized collaborators. The repository also maintains version history to help track updates and ensure accountability for file modifications.
This feature facilitates collaborative development by allowing team members to securely share and manage project files throughout the project lifecycle.
Priority Level: High

5.4.2 Stimulus/Response Sequences

After creating a project, the system automatically generates a repository associated with that project. Authorized collaborators can access the repository through the project page.
Users may upload new files, edit existing files, download files, or view previous versions of uploaded documents. Whenever a file is uploaded or modified, the system updates the repository and records the contributor, timestamp, and version information. Other collaborators can immediately view the latest repository contents.

5.4.3 Functional Requirements

REQ-1: Create Repository
User: Project Owner
Input:

- Create a new project.

Output:

- A repository is automatically created and linked to the project.

Alternative flow(s):

- If repository creation fails due to a server or storage error, the system displays an error message and retries the operation.

REQ-2: Upload Project Files
User: Project Owner and Collaborators
Input:

- Open Repository.
- Click Upload File.
- Select one or more supported files.
- Click Upload.

Output:

- Selected files are uploaded successfully.
- Repository contents are updated.
- Upload details including contributor and timestamp are recorded.

Alternative flow(s):
- Unsupported file type.
- File size exceeds permitted limit.
- Upload interrupted due to network failure.

The system displays an appropriate error message and allows the user to retry.

REQ-3: View and Download Repository Files
User: Project Owner and Collaborators
Input:
- Select a file from the repository.

Output:
- File preview (where applicable).
- Download option.
- Version history.

Alternative flow(s):
- Selected file no longer exists.
- User lacks permission to access the repository.

REQ-4: Update Repository Files
User: Project Owner and Collaborators
Input:
- Select an existing file.
- Upload an updated version or edit supported file types.
- Save changes.

Output:
- Repository is updated.
- Previous version is retained in version history.
- Contributors are notified of the update.

Alternative flow(s):
- Simultaneous modification conflict.
- Storage unavailable.
- User does not have edit permission.

5.5 Messaging and Notifications

5.5.1 Description and Priority

The Messaging and Notifications feature enables users to communicate directly with collaborators and project owners. The system supports one-to-one conversations as well as project group chats for coordinating project activities. Notifications inform users about collaboration requests, invitations, repository updates, new messages, and project changes. Real-time communication ensures efficient collaboration among project members.
Priority Level: High

5.5.2 Stimulus/Response Sequences

A user opens another user's profile or a project chat and sends a message. The system immediately delivers the message to the recipient using real-time communication services.
Whenever a collaboration request is received, accepted, rejected, or when repository files are updated, the system generates notifications and displays them in the user's notification panel. Users may view, read, and clear notifications at any time.

5.5.3 Functional Requirements

REQ-1: Send Direct Message
User: Project Owners and Collaborators
Input:

- Open Chat.
- Select a user.
- Type a message.
- Click Send.

Output:

- Message is delivered to the recipient.
- Conversation history is updated.

Alternative flow(s):

- Recipient is offline.
- Temporary network failure.

The system stores the message and delivers it once connectivity is restored.

REQ-2: Project Group Chat
User: Project Owner and Collaborators
Input:

- Open Project Chat.
- Type message.
- Click Send.

Output:

- Message is visible to all project collaborators.
- Chat history is updated in real time.

Alternative flow(s):

- User is no longer a project collaborator.
- Project chat is unavailable due to server issues.

REQ-3: Receive Notifications
User: Project Owners and Collaborators
Input:
System-generated events including:

- Collaboration request
- Invitation to project
- Accepted or rejected request
- New message
- Repository update
- Collaborator added or removed

Output:
- Notification is displayed in the notification panel.
- Notification count is updated.

Alternative flow(s):
- User is offline.
- Notification service temporarily unavailable.

Notifications are delivered once the user reconnects.

REQ-4: Manage Notifications
User: Project Owners and Collaborators
Input:
- Open Notification Panel.
- Mark notification as read or clear notification.

Output:
- Notification status is updated.

Alternative flow(s):
- Notification has already been deleted.
- Synchronization failure due to network interruption.

5.6 Recommendation System

5.6.1 Description and Priority

The Recommendation System assists users in discovering relevant projects and collaborators based on their skills, interests, technologies, previous collaborations, and profile information. It continuously analyzes user profiles and project requirements to present personalized recommendations on the Home Page. This feature improves project visibility and helps users quickly find suitable collaboration opportunities.
Priority Level: Medium

5.6.2 Stimulus/Response Sequences

When a user logs into the platform, the system retrieves the user's profile, skills, interests, and collaboration history. Based on this information, the recommendation engine identifies projects requiring similar skills and collaborators with complementary expertise. The generated recommendations are displayed on the Home Page and updated whenever relevant profile or project information changes.

5.6.3 Functional Requirements

REQ-1: Recommend Projects
User: Project Owners and Collaborators
Input:
System analyzes:
- User skills
- Areas of interest
- Previous projects
- Technologies known

Output:
Displays a list of recommended projects on the Home Page.
Alternative flow(s):

- Insufficient user profile information.
- No matching projects available.

The system displays recently created projects instead.

REQ-2: Recommend Collaborators
User: Project Owners
Input:
System analyzes:

- Project requirements
- Required technologies
- Open positions
- User skill profiles

Output:
Displays suitable collaborator recommendations for the selected project.
Alternative flow(s):

- No users match the project requirements.
- Required role has already been filled.

REQ-3: Update Recommendations
User: Project Owners and Collaborators
Input:
Changes in:

- User profile
- Skills
- Project requirements
- Collaboration history

Output:
Recommendations are automatically refreshed to reflect the updated information.
Alternative flow(s):

- Recommendation service temporarily unavailable.

Previously generated recommendations continue to be displayed until new recommendations become available.

6 Other Nonfunctional Requirements

6.1 Performance Requirements

CollabCupid should provide a responsive and reliable user experience while supporting real-time collaboration among multiple users. The system must efficiently handle project searches, messaging, repository operations, and profile management with minimal delay.
Performance requirements include:

- Average page load time should not exceed 3 seconds.
- Project and collaborator search results should be displayed within 2 seconds.
- Chat messages and notifications should be delivered in real time (within 500 ms under normal network conditions).
- Repository pages should load within 3 seconds, and file uploads/downloads should begin promptly.
- The system should support at least 100 concurrent users without significant performance degradation.
- Frequently accessed data should be optimized using efficient database queries and caching techniques.
- The application should be scalable to accommodate increasing numbers of users, projects, repositories, and messages.
- System availability should be at least 99%, with regular backups to prevent data loss.
- Client and server resources should be efficiently utilized to ensure smooth performance across supported devices and browsers.

6.2 Safety Requirements

CollabCupid stores user profiles, project repositories, collaboration requests, and chat messages. Therefore, appropriate measures must be taken to protect user data and ensure safe communication between collaborators.

- Users should use strong passwords and avoid sharing login credentials with others.
- The system should validate all user inputs to prevent invalid or malicious data from being stored.
- Uploaded files should be scanned and restricted to supported file types to reduce the risk of malicious uploads.
- Collaboration requests and repository access should be restricted to authorized users only.
- Regular backups should be maintained to prevent data loss due to system failures.
- Users should be able to report inappropriate projects, users, or messages to maintain a safe collaborative environment.
- The system should provide confirmation before performing critical actions such as deleting projects, removing collaborators, or permanently deleting repository files.
- In case of unexpected failures, the system should recover gracefully without corrupting user data or project repositories.

6.3 Security Requirements

CollabCupid must ensure the confidentiality, integrity, and availability of user data while preventing unauthorized access to projects and repositories.

- All communication between the client and server shall be secured using HTTPS/SSL encryption.
- User passwords shall be securely hashed and never stored in plain text.
- Authentication shall be implemented using Firebase Authentication, with secure session management.
- Only authenticated users shall be allowed to create projects, send collaboration requests, access repositories, or use the messaging system.
- Authorization for every database read and write shall be enforced by Firestore Security Rules, evaluated on Google's servers independently of the client application.
- Access to project repositories and group chats shall be restricted to the project owner and authorized collaborators through role-based access control.
- User inputs shall be validated and sanitized to prevent common security attacks such as SQL injection and Cross-Site Scripting (XSS).
- File uploads shall be restricted to supported file types and validated before being stored.
- User sessions shall automatically expire after a period of inactivity, requiring users to log in again.
- The system shall maintain activity logs for important actions such as project creation, collaborator addition or removal, and repository updates to improve accountability and security.

6.4 Software Quality Attributes

CollabCupid shall be designed to provide a reliable, scalable, and user-friendly platform for project collaboration. The system should maintain high software quality while supporting future enhancements.

- Usability: The user interface shall be simple, intuitive, and responsive, enabling users to easily search projects, manage repositories, and communicate with collaborators.
- Reliability: The system shall operate consistently with minimal failures, ensuring that user data, chat history, and repository files are not lost.
- Maintainability: The application shall follow a modular architecture, allowing new features and updates to be added with minimal changes to the existing system.
- Scalability: The system shall support increasing numbers of users, projects, repositories, and messages without significant degradation in performance.
- Availability: The platform shall be accessible whenever required, with minimal downtime and regular database backups.
- Portability: The application shall function correctly across major web browsers and devices, including desktops, tablets, and smartphones.
- Performance: The system shall provide fast response times for searches, messaging, and repository operations, ensuring a smooth user experience.
- Interoperability: The application shall integrate seamlessly with external services such as Firebase Authentication, Cloud Firestore, and Firebase Storage while maintaining data consistency.

6.5 Business Rules

Only registered and authenticated users shall be permitted to create projects, join projects, send collaboration requests, or access repositories. A project owner is responsible for creating, editing, and managing the project, including adding or removing collaborators. Users may join a project only after their collaboration request has been approved or they have accepted an invitation from the project owner. Projects with all collaboration positions filled shall no longer appear in the Home Page or search results until a position becomes available. Only authorized collaborators shall have access to the project's repository, shared files, and project group chat.

7 Other Requirements

The system requires a stable internet connection, cloud database services, and secure authentication services such as Firebase or Supabase for proper operation. Regular database backups and cloud storage are required to ensure the availability and integrity of user profiles, repositories, and chat data.
CollabCupid does not process payments and has no payment gateway dependency. A defined privacy policy and SSL certification, provided automatically by Firebase Hosting, protect user and system data.

Appendix A: Glossary

Some key terms used repeatedly throughout the document are

1. **Collaborator -** A registered user who contributes to a project.
2. **Project Owner -** The user who creates and manages a project.
3. **Repository -** A storage space where project files and source code are maintained.
4. **Collaboration Request -** A request sent by a user to join a project or an invitation sent by a project owner.
5. **Open Position -** A role within a project that is available for new collaborators.
6. **Profile -** A user's public information, including skills, interests, and previous projects.
7. **Authentication -** The process of verifying a user's identity before granting access.
8. **Authorization -** The process of determining a user's permissions within the system.
9. **Firestore Security Rules -** A declarative, server-side policy evaluated by Firestore on every read and write, used in place of a backend API.
10. **Real-Time Messaging -** Instant communication between users via Cloud Firestore's real-time listeners (onSnapshot).
11. **Notification -** A system-generated alert informing users about collaboration requests, messages, or project updates.
12. **Version History -** A record of changes made to files in a project repository.
13. **Role-Based Access Control (RBAC) -** A security mechanism that grants permissions based on user roles such as Project Owner or Collaborator.
14. **Cloud Storage -** Online storage used to securely store repository files and user uploads.
15. **Home Page -** The main page displaying recommended projects and navigation options after login.